

Grade 7 Accelerated
Unit 3
Lesson 2 Practice

Name Answer Key
Date _____
Period _____

Show your steps of applying the quotient of powers exponent law to evaluate.

1. $\frac{10^5}{10^3}$ $10^{5-3} = 10^2 = 10 \cdot 10 = 100$

2. $\frac{8^4}{8^1}$ $8^{4-1} = 8^3 = 8 \cdot 8 \cdot 8 = 512$

3. $\frac{(-6)^6}{(-6)^5}$ $(-6)^{6-5} = (-6)^1 = -6$

4. When dividing two powers that have the same nonzero base, keep the base and subtract the exponents. This is called the quotient of powers exponent law.

Apply the power of quotient exponent law. Do not simplify the fraction.

5. $\left(\frac{-5}{10}\right)^7$ $\frac{(-5)^7}{10^7}$

6. Fill in the missing value in the table.

exponent form	4^4	4^3	4^2	4^1	4^0
value	256	64	16	4	1

7. Evaluate 9^0

$$9^0 = 1$$

Rewrite the expression by using the laws of exponents. Circle the exponent law used from the list provided. Then, evaluate the exponential expression.

8.

Expression	Exponent Law	Evaluated Expression
$\frac{7.6^5}{7.6^3}$	• Quotient of Powers Law • Power of a Quotient Law • Zero Exponent Law	$7.6^{5-3} = 7.6^2 = 7.6 \cdot 7.6 = 57.76$

9.

Expression	Exponent Law	Evaluated Expression
$\left(\frac{10}{11}\right)^0$	• Quotient of Powers Law • Power of a Quotient Law • Zero Exponent Law	$\frac{10^0}{11^0} = 1$

10.

Expression	Exponent Law	Evaluated Expression
$\left(\frac{0.8}{1.6}\right)^4$	• Quotient of Powers Law • Power of a Quotient Law • Zero Exponent Law	$\frac{(0.8)^4}{1.6^4} = \frac{0.4096}{6.5536} = 0.0625$ or $0.8 \div 1.6 = 0.5 \quad (0.5)^4 = 0.0625$